

27 - Group 2

solubilities of Group 2 sulfates & hydroxides:

$$\Delta H_{\text{hyd}} = \Delta H_{\text{latt}} + \Delta H_{\text{sol}}$$

more energy is released to break the lattice

$$\begin{matrix} \Delta H_{\text{sol}} \\ \text{more exothermic} \\ \text{more solubility} \end{matrix} = \begin{matrix} \Delta H_{\text{hyd}} \\ \text{bigger} \end{matrix} - \begin{matrix} \Delta H_{\text{latt}} \\ \text{smaller} \end{matrix}$$

$$\Delta H_{\text{sol}} = \Delta H_{\text{hyd}} - \Delta H_{\text{latt}}$$

$$\begin{matrix} \Delta H_{\text{sol}} \\ \text{less exothermic} \\ \text{less solubility} \end{matrix} = \begin{matrix} \Delta H_{\text{hyd}} \\ \text{smaller} \end{matrix} - \begin{matrix} \Delta H_{\text{latt}} \\ \text{bigger} \end{matrix}$$

solubilities of Group 2 sulfates

★ hypothetical values

	$\Delta H_{\text{hyd}} (M^{2+})$	ΔH_{latt}	ΔH_{sol}	
MgSO ₄	-3600	-2000	$(-3600) - (-2000) = -1600$ most exo	exothermic nature decreases solubility decreases
CaSO ₄	-2800	-1800	$(-2800) - (-1800) = -1000$	
SrSO ₄	-2000	-1600	$(-2000) - (-1600) = -400$	
BaSO ₄	-1200	-1400	$(-1200) - (-1400) = +200$ least exo (endo)	

* size of M^{2+} ion increases down the group → less charge density → less ΔH_{hyd} & ΔH_{latt}

* values of ΔH_{hyd} decrease more steeply compared to ΔH_{latt}

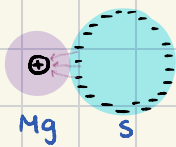
solubilities of Group 2 hydroxides

	$\Delta H_{\text{hyd}} (M^{2+})$	ΔH_{latt}	ΔH_{sol}	
Mg(OH) ₂	-3600	-3000	$(-3600) - (-3000) = -600$ least exo	exothermic nature increases solubility increases
Ca(OH) ₂	-3300	-2400	$(-3300) - (-2400) = -900$	
Sr(OH) ₂	-3000	-1800	$(-3000) - (-1800) = -1200$	
Ba(OH) ₂	-2700	-1200	$(-2700) - (-1200) = -1500$ most exo	

* size of M^{2+} ion increases down the group → less charge density → less ΔH_{hyd} & ΔH_{latt}

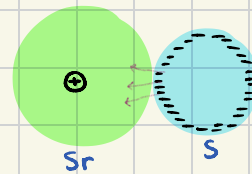
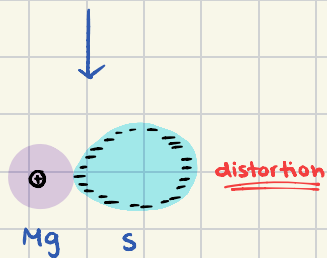
* values of ΔH_{latt} decrease more steeply compared to ΔH_{hyd}

Ion polarisation

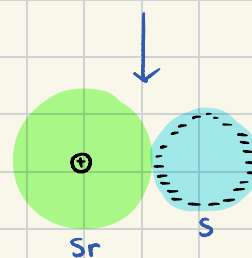


electrostatic attraction b/w the nucleus of the cation & electron cloud of anion

Mg²⁺ ion pulls the e⁻ cloud of S²⁻ ion strongly



Sr²⁺ ion cannot pull the e⁻ cloud of S²⁻ ion as strongly



Ion polarisation

The distortion of the e⁻ cloud of an anion under the pull of the cation

⇒ polarising power

the ability of cation to distort the e⁻ cloud of anion so that it's no longer spherical

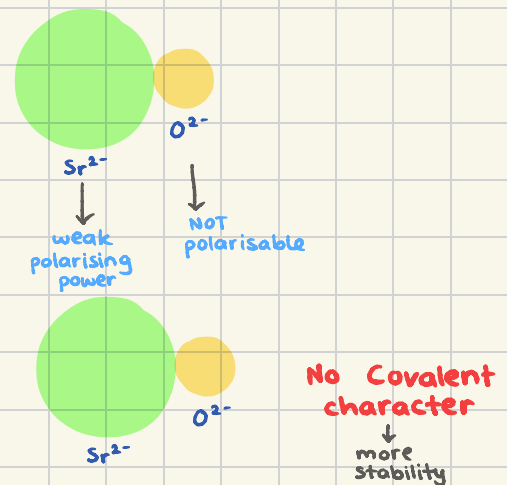
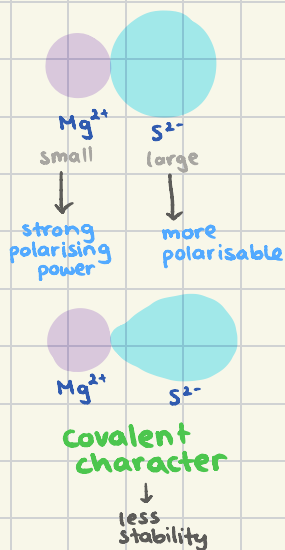
smaller & highly charged cations
= more polarising power

⇒ polarisability

the ability of anion to undergo distortion by the effect of cation

Larger & small-charged anions
= more polarisability

e.g



Stability of group 2 compounds

